MILITARY SPECIFICATION

BREATHING APPARATUS, OXYGEN GENERATING, M20

1. SCOPE

1.1 This specification covers one type of self-contained oxygen-generating breathing apparatus, which is back mounted.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

NN-P-00515	-	Plywood, Container Grade.
CCC-T-191	-	Textile Test Methods.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-636	-	Boxes, Fiberboard.
PPP-B-665	-	Boxes; Paperboard, Metal Stayed (Including Stay Material).

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-A-13550	-	Antifogging Kit, M1.
MIL-C-17671	-	Canister, Oxygen Breathing Apparatus.

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STANDARDS

FEDERAL

FED-STD-595	-	Colors.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking for US Military Property.
MIL-STD-289	-	Visual Inspection Guide for Rubber Sheet Material.
MIL-STD-407	-	Visual Inspection Guide for Rubber Molded Items.

PUBLICATIONS

DEPARTMENT OF THE ARMY TECHNICAL MANUALS

TM 3-4240-212-15	-	Operator, Organizational, Field, and Depot Maintenance Manual for Breathing Apparatus, Oxygen Generating, M20.
TM 3-4240-212-25P	-	Organizational, Field, and Depot Maintenance Repair Parts and Special Tool List for Breathing Apparatus, Oxygen Generating, M20.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules, and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

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3. REQUIREMENTS

3.1 Design. The apparatus shall be a permissible, one hour type, self-contained, oxygen-generating, breathing apparatus, US Bureau of Mines Approval No. 1307, modified as follows (see 6.3):

3.1.1 Speech diaphragm. The apparatus shall not have a speech diaphragm.

3.1.2 Cross tube. The cross tube shall be turned toward the back and upward at an angle of 60° ± 5° and 30° to 35°, respectively. The turned back portion shall start 1-1/2 ± 1/2 inch from the center of the tube extended from the facepiece. The turned back portion shall be 2-1/4 ± 1/4 inch long.

3.1.3 Inlet and outlet valves. The inlet and outlet valves shall be reversed so that the inlet valve is on the right-hand side of the wearer and the outlet valve shall be on the left-hand side of the wearer.

3.1.4 Pressure relief valve. A spring-loaded pressure relief valve shall be built into the right-hand side turned back portion of the cross tube. The valve shall be turned in toward the facepiece at 40° to 45° from the vertical of the turned back end of the cross tube when the cross tube ends are in a horizontal position. The valve shall be spring loaded and it shall begin to open at no less than 3 pounds applied load and shall be fully opened with an applied load of 13 ± 2 pounds when tested as specified in 4.3.4.1(a). The valve shall show no evidence of leakage when subjected to a pressure equal to 6 + 0.5 inch of water for 60 ± 15 seconds when tested as specified in 4.3.4.1(b).

3.1.5 Breathing tubes. The breathing tubes shall be 20 ± 1 inch long.

3.1.6 Back-carry harness assembly. The apparatus shall be designed for back carry instead of chest carry. The apparatus shall be provided with an adjustable-fit harness assembly made of polyvinylidine chloride (saran) plastic webbing, 2 ± 1/4 inch wide for shoulder and chest straps and 1 ± 1/8 inch for other straps, for fastening the apparatus securely to the wearer's back without discomfort. All harness stitching shall be polyester fiber (dacron) stitching. The harness shall permit freedom of motion for physical work. The harness material joints shall have a breaking strength of no less than 150 pounds for 2-inch-wide straps, when tested as specified in 4.3.4.2. A cushion pad of approximately 8 inches by 4 inches shall be placed on the lower portion of the unit so that it rests against the small of the back of the wearer.

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3.1.7 Carrying case design and construction. A carrying case shall be provided to retain the apparatus, a spanner wrench to couple and uncouple the hoses, two type I canisters and one type II canister (see 3.1.9), instruction booklet or card, an antifogging kit (see 3.1.11) and a hood (see 3.1.12). Unless otherwise specified in the contract or order, the color of the carrying case shall be green No. 34087 of FED-STD-595. If the case is provided with any metal parts, they shall be a corrosion resistant metal or metal parts treated with standard commercial treatment for corrosion resistance prior to painting.

3.1.8 Spanner wrench. A spanner wrench shall be provided to couple and uncouple the breathing tubes.

3.1.9 Canisters. Each apparatus shall be provided with two type I canisters and one type II canister conforming to MIL-C-17671.

3.1.10 Instructions. One copy each of TM 3-4240-212-15 and TM 3-4240-212-25P shall be included in the carrying case. A copy of the manufacturer's instructions shall be included in the carrying case.

3.1.10.1 Changes in design. Prior to the initiation of production, the contractor shall advise the Government of any changes that affect operating and maintenance instructions or repair parts and special tools list which are in effect at the time of procurement.

3.1.11 Antifogging kit, M1. An antifogging kit conforming to MIL-A-13550 shall be packed into the carrying case.

3.1.12 Hood. One hood supplied as an accessory shall be Mine Safety Appliance Company part No. 83461 (see 6.4).

3.2 Breathing bag leakage. The breathing bag shall show no evidence of leakage when subjected to a pressure equal to 10 + 0.5 inch of water for 60 ± 15 seconds when tested as specified in 4.3.4.3.

3.3 Functioning. The apparatus shall pass the functioning test as specified in 4.3.4.4.

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3.4 Identification marking.

3.4.1 Breathing apparatus. Identification marking of the apparatus shall be in accordance with the applicable requirements of MIL-STD-130. The use of a nameplate, properly attached, shall be permitted.

3.4.2 Carrying case. Identification marking prescribed in 3.4.1 shall be stenciled on the carrying case lid. Alternatively, a nameplate shall be permitted.

3.5 Material. The rubber parts used in the assembly of the breathing apparatus shall have been manufactured within six months of the date of assembly.

3.6 Preproduction. Prior to the start of regular production, a preproduction sample of breathing apparatus and harness joints shall be produced in accordance with this specification for examination and test (see 4.2).

3.7 Workmanship. The apparatus shall be free from contamination (grease or oil) and damage (dents, scratches, or punctures).

3.7.1 Rubber surfaces. The bag, hoses, and facepiece, shall be free from tackiness, porous areas, cracks, checks, flash, abrasions, and blisters. The rubber shall be compact and free from bubbles and foreign matter. Ridges of moldparting lines, when closely trimmed, shall not be considered as flash.

3.7.2 Metal surfaces. The metal parts shall have no sharp edges which might conceivably injure a wearer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

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4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Preproduction.

4.2.1 Sample. A preproduction sample of 5 M20 breathing apparatus and 5 harness joint specimens, 6 inches by full width, shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.2.2 Inspection.

4.2.2.1 For examination. Sample apparatus shall be examined in accordance with the classification of defects (4.3.3.3).

4.2.2.2 For tests. Sample apparatus and harness joint specimens shall be tested as specified in 4.3.4.

4.2.3 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified in 4.2.2 to be acceptable.

4.3 Regular production.

4.3.1 Lotting. A lot shall consist of the apparatus or component parts produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions.

4.3.2 Sampling.

4.3.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.3.2.2 For destructive tests. A sample of 3 harness joint specimens, six inches by full width, shall be made essentially the same as those in the apparatus and submitted with each lot of apparatus for testing.

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4.3.3 Inspection.

4.3.3.1 Examination and nondestructive tests. Sample apparatus and the level A preparation for delivery shall be examined and tested in accordance with the classification of defects and MIL-STD-105. MIL-STD-289 and MIL-STD-407 shall apply as visual inspection guides for rubber sheet material and rubber molded items.

4.3.3.2 Destructive tests. Sample harness joint specimens shall be tested in accordance with 4.3.4.2. The acceptance number is zero.

4.3.3.3 Classification of defects.

(a) Breathing apparatus, oxygen generating, M20.

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 0.25 percent defective	
101	Relief valve spring load	4.3.4.1(a)
102	Relief valve leakage	4.3.4.1(b)
103	Breathing bag leakage	4.3.4.3
104	Functioning	4.3.4.4
	AQL 1.0 percent defective	
105	Component incorrectly located	
106	Type or quantity of canisters incorrect	
107	Marking missing, incorrect or illegible	
108	Manufacturer's instructions missing	
109	Workmanship (see 3.7)	
Minor:	AQL 2.5 percent defective	
201	Spanner wrench missing	
202	Antifogging kit missing	
203	Hood missing	
204	Contamination on interior or exterior of carrying case	
205	Color of carrying case incorrect	

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(b) Preparation for delivery (see section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Container incorrect or damaged
102	Closure of carrying case or container incorrect
103	Packaging method incorrect
104	Cushioning, blocking, or bracing missing or insufficient to make a tight package or pack
105	Marking missing, incorrect, or illegible

4.3.4 Tests. Tests shall be conducted as follows:

4.3.4.1 Pressure relief valve.

(a) Spring load. Use a calibrated spring tester adapted for the apparatus and gradually apply a load to the pressure relief valve. Record the load at which the valve begins to open and the load at which the valve is fully opened.

(b) Leakage. Assemble the pressure relief valve to a suitable fixture with an adjustable oxygen supply and a water manometer making certain all connections are airtight. Adjust the oxygen supply to obtain a reading of 6 + 0.5 inch on the water manometer and close off the oxygen supply. The pressure relief valve shall show no evidence of pressure drop for 60 ± 15 seconds as indicated by the manometer.

4.3.4.2 Strength of harness joint. Determine the strength of the harness joint in accordance with method 5100 of CCC-T-191, except that the test specimen shall be full width by 6 inches long.

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4.3.4.3 Breathing bag leakage. Disconnect the two breathing tubes from the breathing bag assembly and install a rubber stopper in the inhalation elbow. Install and seat a dummy canister so that it makes an airtight connection. Connect a suitable oxygen source and a water manometer to the top of plunger valve assembly on the breathing bag assuring that all connections are airtight. Adjust the oxygen source to inflate the breathing bag to a pressure equal to 12 inches of water as measured on the manometer. While inflating, straighten out the breathing bag by kneading and pulling the bag to a fully inflated position. Reduce the pressure to obtain a reading of 10 + 0.5 inch of water on the manometer and close off the oxygen source. The breathing bag shall show no evidence of pressure drop for 60 ± 15 seconds as indicated by the manometer. Disconnect the oxygen source, remove the rubber stopper and the dummy canister.

4.3.4.4 Functioning. Don the facepiece and hoses, adjust the straps if necessary so that the facepiece fits snugly and comfortably against the face. Cover the coupling at the end of the hose on the wearer's right-hand side with the palm of the hand and inhale; the facepiece shall collapse. Cover the coupling at the end of the hose on the wearer's left-hand side with the palm of the hand and exhale; the facepiece shall inflate and the exhaled air shall escape around the edges of the facepiece. Connect the breathing tubes to the breathing bag assembly.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level A.

5.1.1 Preservation and packaging methods. Preservative compounds or oils shall not be applied to any surface of the breathing apparatus. Packaging methods cited herein shall be in conformance with the requirements of MIL-P-116.

5.1.1.1 Timer. The timer shall be removed from the breathing apparatus and packaged in accordance with method IIe, in either a paperboard folding box conforming to PPP-B-566; a style A, paperboard box conforming to PPP-B-665; or a type CF, class domestic fiberboard box conforming to PPP-B-636.

5.1.1.2 Technical publications. The technical publications shall be packaged in accordance with method IC-3.

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5.1.1.3 Breathing apparatus. The breathing apparatus, consisting of a facepiece with hose assembly, back-carry harness assembly, breathing bag assembly, antifogging kit, hood, canisters, and spanner wrench together with the packaged timer and technical publications shall be placed inside the carrying case (see 3.1.7) in such a manner as to prevent any permanent set in the rubber components of the apparatus. Internal cushioning, blocking, or bracing shall be employed as required to prevent movement of the apparatus or components within the carrying case. The cover shall be closed and secured with the fasteners provided.

5.2 Packing. Level shall be as specified (see 6.2).

5.2.1 Level A. The breathing apparatus, packaged as specified in 5.1, shall be packed in an overseas-type, style A, cleated-plywood box, conforming to PPP-B-601 and using type II, class 2 plywood conforming to NN-P-00515. The box shall be closed and strapped in accordance with the appendix to PPP-B-601. Galvanized strapping shall be used.

5.2.2 Level B. The breathing apparatus, packaged as specified in 5.1, shall be packed in a V3c, style RSC, fiberboard box, conforming to PPP-B-636. Internal fiberboard blocking shall be provided around chest hardware to prevent the hardware from abrading or rupturing the fiberboard box. The fiberboard box shall then be closed and reinforced in accordance with the appendix of PPP-B-636, for exterior boxes, using metal or tape banding and sealing the box seams and manufacturer's joint as specified therein. When metal banding is used as a reinforcement, the banding shall be galvanized.

5.2.3 Level C. The breathing apparatus, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Uniform Freight Classification rules or the regulations of other common carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.4 Repair parts. When this item is being procured for stockage and issue as a repair part, preservation, packaging, packing and marking shall be as specified on the repair parts packaging data sheet (see 6.5).

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6. NOTES

6.1 Intended use. The breathing apparatus is intended for use where a self-contained oxygen-generating breathing apparatus is required.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packing required (see 5.2)
(c) Serial number required
(d) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.

(2) Name and address of test facility and shipping instructions when testing is performed by the Government.

(3) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Source. The breathing apparatus covered by this specification is Mine Safety Appliance Company Catalogue Number 81812, Federal Stock Number 4240-678-5263. This apparatus is a revised "Chemox" mask.

6.4 Hood. The Federal Stock Number for the hood is 4240-713-2499.

6.5 Packaging data sheet. The packaging data sheet is identified by the Federal Stock Number for this item and should be ordered from the Commanding Officer, Edgewood Arsenal, ATTN: SMUEA-TSE-DD, Edgewood Arsenal, Maryland 21010.

6.6 Supersession data. This specification supersedes Edgewood Arsenal Purchase Description 197-54-702A(1), dated 28 November 1962.

Custodian:

Army - MU

Preparing activity:

Army - MU (EA)

Project No. 4240-A268

*U.S. GOVERNMENT PRINTING OFFICE: 1967-251-519/5674
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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119 R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

TO DETACH FORM - CUT ON THIS LINE

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